Developing AILA: An Intelligent Lecturing Assistant

From Research Pipeline to Full-Stack Platform with Adaptive Assessment

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STAR SCHOLARS

UREP Mini-Grant $1,000

Pennoni Honors College Student Showcase

MOTIVATION & PROBLEM

How can AI automate high-quality formative assessment from lecture materials — and adapt to individual student mastery?

Instructors in fast-moving STEM fields face two compounding challenges: (1) manually writing high-quality retrieval practice questions is labor-intensive, and (2) AI-generated questions often suffer from hallucinations and weak distractors, instructors filter up to two-thirds of output before use. AILA addresses both by automating the full pipeline from lecture upload to adaptive quiz delivery.

PROJECT JOURNEY

2025 STAR Scholars Program

Built the KG-based MCQ pipeline. Evaluated 400+ questions across 14 quality criteria.

2025 IEEE BigData Published

Rate-Distortion Guided KG Construction. Macau SAR, Dec 2025. arXiv:2511.14595

2026 UREP Mini-Grant ($1,000)

Received $1,000 research grant. Transitioned from a research script to a full-stack deployable web app.

2026 ACM SIGCSE Published

Scaling Retrieval Practice with LLM. St. Louis, MO. DOI: 10.1145/3770761.3777185

2026 Independent Study (Now)

Benchmarking graph-based CAT vs. linear IRT using DINA models on the AILA platform.

PUBLISHED RESEARCH

IEEE BigData 2025 · arXiv:2511.14595

Rate-Distortion Guided KG Construction from Lecture Notes Using Gromov-Wasserstein Optimal Transport

Formalizes KG extraction as information-theoretic compression. FGW transport measures structural + semantic distortion. Five operators minimize L = R + βD.

ACM SIGCSE TS 2026 · DOI: 10.1145/3770761.3777185

Scaling Retrieval Practice with LLM: Improving MCQ Quality through Knowledge Graphs

257 KG-based vs. 207 Raw MCQs, rated on 14 Likert criteria. KG MCQs score +0.52 on 'Hard to Guess' — the strongest gain across all criteria.

arXiv:2507.05629 · Co-authored with Dr. Yuan An et al.

Enhancing Student Learning with LLM-Generated Retrieval Practice Questions: An Empirical Study

10-week study: 89% accuracy with LLM practice vs. 73% control — a 16-point gain (p < 0.0001, r = 0.586). Co-authored during AILA's development phase.

AILA PLATFORM — SYSTEM ARCHITECTURE

Next.js / React	FastAPI (Python)	Google Gemini LLM	SQLite
Frontend	Backend	AI Model	Database

1 Upload Lecture — PDF/PPTX uploaded. Text extracted per-slide.

2 Build KG — Two-pass Gemini pipeline builds knowledge graph.

3 Generate MCQs — KG-guided prompts produce higher-order MCQs.

4 Adaptive Quiz — Students receive difficulty-configured quizzes.

KG-BASED VS. RAW MCQ — METHODOLOGY

	Raw MCQ	KG-Based MCQ
Context	Current slide only	Full lecture knowledge graph
Input	Raw slide text	KG concepts + edges (JSON)
Focus	Fact recall	Conceptual, cross-topic
Difficulty	Single-slide	Higher-order, integrative

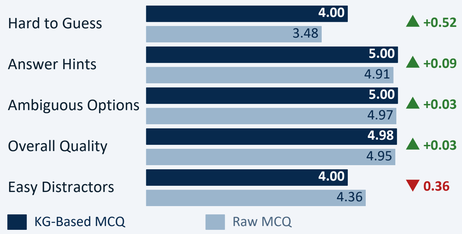
EMPIRICAL RESULTS — PUBLISHED RESEARCH

89%	73%	+16pt	60+
Accuracy with AI practice	Accuracy control group	Improvement p < 0.0001	Students in study

Source: arXiv:2507.05629 (co-authored). Mann-Whitney U, effect size r = 0.586.

MCQ QUALITY — 14-CRITERION LIKERT STUDY

257 KG-based vs. 207 Raw MCQs — CS172 Drexel. Likert 1–5.



RESEARCH IMPACT

For Instructors

Hours → seconds for question creation. KG MCQs rated higher-quality by instructors.

For Students

16-point accuracy gain (p < 0.0001). Instant feedback reinforces concept mastery.

For Research

Open platform for adaptive assessment at scale. Ongoing IS: graph-based CAT vs. IRT.

KNOWLEDGE GRAPH — RATE-DISTORTION FRAMEWORK

KG extraction is formalized as an information-theoretic compression problem.
Fused Gromov-Wasserstein (FGW) transport quantifies structural + semantic distortion. Iterative refinement minimize L = R + βD.

R — Rate: KG size / complexity. More nodes = more information encoded.

D — Distortion: Semantic + structural divergence via FGW coupling.

L — Lagrangian: L = R + βD. Minimized by: add, merge, split, prune, rewire.

KG Refinement Operations:

Add concept | Merge similar | Split coarse | Prune redundant | Rewire edges

TWO-PASS KG GENERATION PIPELINE

Pass 1 — Structure

Reads all slides in one prompt
Identifies topics + depth scores
Extracts inter-topic links

Pass 2 — Concepts

Sequential per-subtopic calls
Extracts child concepts nodes + relationships

AILA WEB PLATFORM — KEY FEATURES

Instructor View

Upload PDF/PPTX lecture slides
View auto-generated KG
Configure quiz settings & difficulty

Student View

Access weekly adaptive quizzes
Answer KG-grounded MCQs
Receive immediate feedback

KEY FINDINGS & ANALYSIS

✓ KG MCQs score +0.52 on 'Hard to Guess' — strongest gain across all 14 criteria.

✓ Overall: KG avg 4.78 vs. Raw 4.75 across 400+ MCQ.

✓ Published: 89% vs. 73% accuracy — 16-pt gain (p < 0.0001, r = 0.586).

△ Limit: 'Easy Distractors' KG −0.36 vs. Raw. Fix: contrast-pair distractor generation.

→ Use Raw MCQs for per-slide recall; KG MCQs for cross-topic integrative exam-prep.

→ Ongoing: IS benchmarks DINA CAT vs. linear IRT — testing KG traversal achieving equivalent diagnostic accuracy

CURRENT WORK — INDEPENDENT STUDY

Benchmarking Graph-Based Computerized Adaptive Testing CAT vs. Linear Item Response Theory (IRT)

Utilizing AILA, benchmarks graph-based CAT against traditional linear IRT. Maps Cognitive Diagnosis Models (DINA) to AILA's KG structure, algorithms evaluated to optimize diagnostic efficiency and minimize question count while maximizing precision in student mastery estimation.

Does graph-based CAT with DINA outperform linear IRT in diagnostic efficiency?

DINA Model

Deterministic Inputs, Noisy 'And' Gate (DINA) model maps concept slip & guess parameters to KG nodes, enabling fine-grained diagnostic precision at concept level.

Graph-Based CAT

Computerized Adaptive Testing traverses KG, selects MCQ based on concept prerequisites & the student's dynamically updated knowledge state.

IRT Benchmark

2PL/3PL IRT models are comparison baseline, measure question count to reach equivalent diagnostic certainty, classification accuracy, & student experience scores.

Adaptive Selection

At each step, algorithm selects MCQ that maximally reduces uncertainty in the student's mastery vector — minimizing test length while preserving diagnostic accuracy.

FUTURE DIRECTIONS

Personalized MCQ generation adapting to mastery in real time

Psychometric IRT calibration to auto-set MCQ difficulty

Instructor review & curriculum alignment tools

Long-term retention studies across multiple terms

Multi-subject expansion: STEM, data science, life sciences

Analytics dashboards with concept mastery heatmaps

CONCLUSIONS

KG MCQs produce more conceptual questions — +0.52 on 'Hard to Guess', avg 4.78 vs. 4.75 Raw.

AI retrieval practice delivers significant learning gains (published, 16-pt improvement) with automated. scalable tooling.

AILA transitions from research pipeline to live full-stack deployable platform.

Ongoing Independent Study: graph-based CAT with DINA vs. linear IRT for adaptive mastery testing.

ACKNOWLEDGEMENTS

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Scan to view the AILA repository

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